

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	3

Case Study Topic: Microsoft Teams Meeting Graph using Strongly Connected Components

Word Done Summary:

In this case study, we analysed Microsoft Teams meeting interactions using graph theory concepts. Meetings were represented as vertices and meeting transitions were represented as directed edges. We applied Tarjan's Strongly Connected Components (SCC) algorithm to identify tightly connected meeting clusters. We also examined the role of discovery time, low-link values, stacks, and DFS traversal in detecting SCCs efficiently.

Implementation Details:

1. Represented meetings as graph vertices.
2. Added directed edges between meetings.
3. Applied Tarjan's SCC algorithm using DFS traversal.
4. Maintained discovery array, low-link array, and stack.
5. Identified strongly connected meeting clusters.
6. Compared Tarjan's algorithm with Kosaraju's algorithm.

Result:

Tarjan's algorithm successfully identified tightly connected meeting clusters in the directed graph. The algorithm efficiently detected strongly connected components using a single DFS traversal with linear time complexity.

Output:

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Strongly Connected Components:  
SCC: m8 m7  
SCC: m6 m5 m4  
SCC: m3 m2 m1
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GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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